Project Design Phase-II Technology Stack (Architecture & Stack)

Date	5 March 2026
Team ID	
Project Name	SB Stocks – Stock Trading Simulation Platform
Maximum Marks	4 Marks

Technical Architecture

The Stock Trading Application follows the MERN Stack (MongoDB, Express.js, React.js, Node.js). The system uses a three-tier architecture consisting of a frontend client, backend API server, and database layer. Users interact through a React web interface, requests are processed by Node.js and Express APIs, and data such as user portfolios and transactions are stored in MongoDB.

The application integrates Stock Market APIs to fetch real-time stock prices and market data.

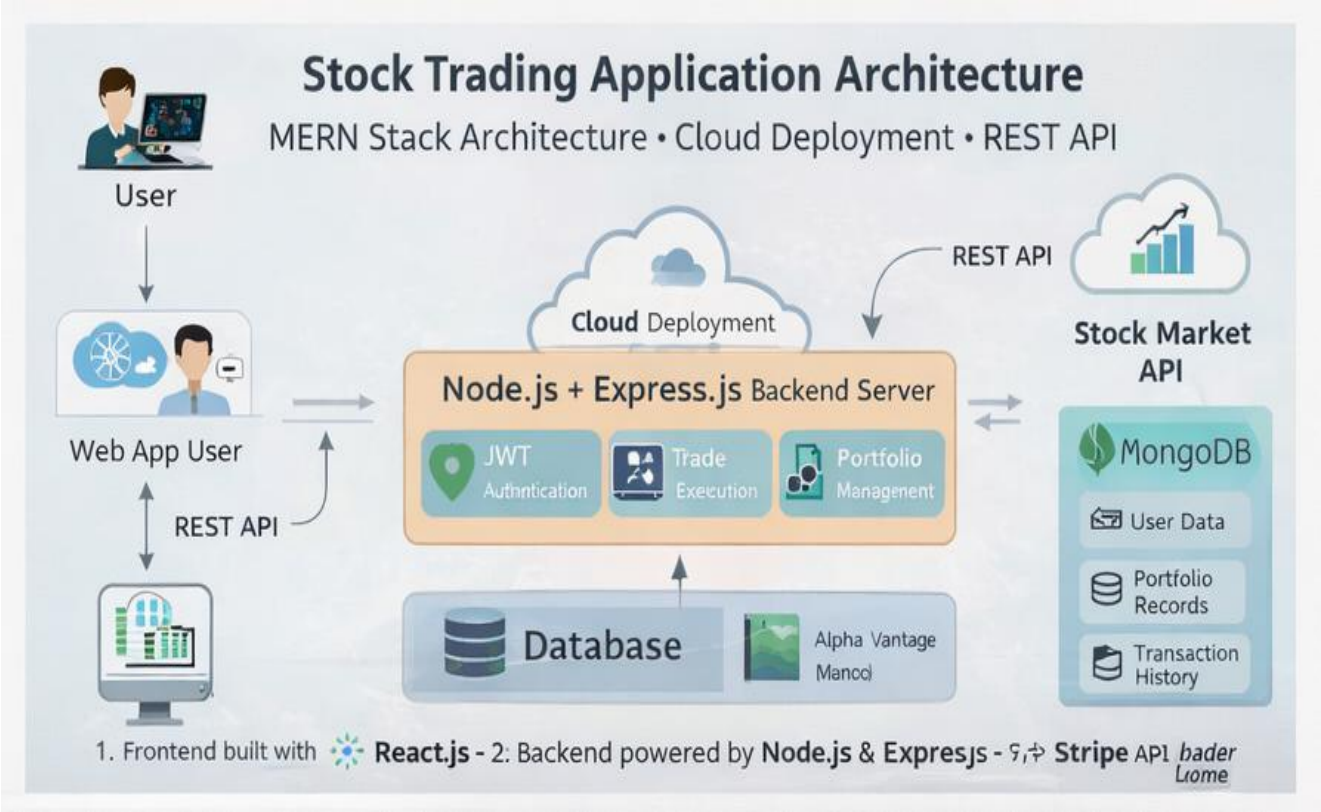

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Allows users to interact with the trading platform	HTML, CSS, JavaScript, React.js
2	Application Logic-1	User authentication and account management	Node.js, Express.js
3	Application Logic-2	Stock buy/sell order processing	Node.js APIs
4	Application Logic-3	Fetch real-time stock market data	Stock Market API
5	Database	Stores user accounts and transaction data	MongoDB
6	Cloud Database	Cloud hosted database service	MongoDB Atlas
7	File Storage	Stores user files and profile images	Cloud Storage / Local Storage
8	External API-1	Real-time stock price data	Alpha Vantage API
9	External API-2	Payment integration	Stripe / Razorpay
10	Machine Learning Model	Stock prediction and analysis	Python ML / TensorFlow
11	Infrastructure	Application deployment and hosting	AWS / Docker

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open Source Frameworks	Frameworks used in development	React.js, Node.js, Express.js, MongoDB

2	Security Implementations	Security and access control mechanisms	JWT Authentication, HTTPS, bcrypt
3	Scalable Architecture	Architecture supports large number of users	Microservices-ready, Docker
4	Availability	Ensures system availability	Cloud Hosting, Load Balancer
5	Performance	Improves response speed and performance	Caching, CDN